Your grade: 92.66%

Your latest: **92.66%** • Your highest: **92.66%** • To pass you need at least 50%. We keep your highest score.

Next item →

1. Select the scenarios where using dropout regularization would be beneficial. **0.8 / 1 point**

- [x] Training a neural network on a small dataset

  Correct
  Correct! Dropout regularization is especially useful when the dataset is small and overfitting is a concern.

- [ ] Training a neural network on a very large dataset
- [x] When the neural network is showing signs of overfitting

  Correct
  Yes! Dropout regularization can help to combat overfitting in neural networks.

- [ ] In reinforcement learning tasks
- [ ] When the model has very few parameters

  You didn't select all the correct answers

2. Which of the following statements are true regarding the impact of dropout regularization on model accuracy? **1 / 1 point**

- [x] It increases training time

  Correct
  Right! Dropout can increase the training time as it requires more epochs to converge.

- [x] It improves test accuracy by preventing co-adaptation of neurons

  Correct
  Well done! Dropout improves test accuracy by preventing neurons from co-adapting.

- [ ] It ensures that the model performs better on the training data
- [ ] It automatically adjusts the learning rate
- [x] It helps prevent overfitting

  Correct
  Correct! Dropout regularization helps in reducing overfitting.

3. How does dropout regularization help in preventing overfitting in neural networks? **1 / 1 point**

- ○ By adding noise to the input data
- ○ By increasing the number of layers in the network
- ◉ By randomly deactivating neurons during training
- ○ By reducing the learning rate

  Correct
  Correct! Dropout works by randomly deactivating neurons during training, which helps in preventing overfitting.

4. How does image data augmentation help improve model accuracy? **1 / 1 point**

- ○ By decreasing the number of training epochs required
- ○ By ensuring the model only sees high-quality images
- ◉ By generating diverse samples, reducing overfitting
- ○ By training the model on the entire dataset multiple times

  Correct
  Correct! Data augmentation helps in creating varied data, which reduces the chances of overfitting.

5. Which of the following are key image data augmentation techniques? **0.8 / 1 point**

- [x] Width shift

  Correct
  Correct! Width shift moves the image horizontally.

- [ ] Blurring
- [x] Cropping

  This should not be selected
  Incorrect. Cropping isn't usually considered an augmentation technique in this context.

- [x] Horizontal flip

  Correct
  Correct! Horizontal flip creates a mirror image, adding diversity to the dataset.

- [x] Rotation

  Correct
  Correct! Rotation alters the angle of images, creating new perspectives.

- [x] Rescaling

  Correct
  Correct! Rescaling is a fundamental augmentation technique.